

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination 2025-26 (NEP)

DEPARTMENT OF STATISTICS**Paper Code: STAMJ41 / STAMN41 — Statistical Inference-I**

Time: 2:30 Hours

Maximum Marks: 70

*Instructions: Attempt **FIVE** questions. Question number one is **compulsory**. Each question carries equal marks (14 marks each).*

Question 1 (Compulsory)

- (a) Discuss consistency criterion of a good estimator. If $X_1, X_2, \dots, X_n$ are n random observations taken from geometric distribution with parameter θ , then show that arX is consistent estimator of $\frac{1}{\theta}$. Also, obtain a consistent estimator of $e^{1/\theta}$. [3.5]
- (b) Explain point and interval estimation with proper examples. Write the confidence interval for the population mean when variance is known. [3.5]
- (c) Discuss sampling distribution of a statistic and standard error with its utility. A machine produces a large number of items of which 15% are found to be defective. If a random sample of 200 items is taken from the population, then find the standard error of sampling distribution of proportion. [3.5]
- (d) Write difference between parametric and non-parametric tests. The cars that enter in a parking are classified either Foreign-made (**F**) or Indian-Made (**I**). To check that the car enter in the parking are in random order, a sample of first 40 cars that entered in the parking is taken and found the following sequence:

I I F F F I F F F F I I F I I I F F I F F I F F I F F I I F F I F F I I

Obtain number of runs and expected value of number of runs in the above sequence? [3.5]

Question 2

- (a) Define Efficiency. If $X_1, X_2, \dots, X_n$ is a random sample taken from a population having mean μ and variance σ^2 , then show that the statistic $T' = \frac{1}{n+1} \sum_{i=1}^n X_i$ is biased but more efficient than sample mean for estimating the population mean. [7]
- (b) If T_1 and T_2 be the two unbiased estimators of $\gamma(\theta)$ with variances σ_1^2 and σ_2^2 and correlation coefficient ρ , what is the best linear unbiased combination of T_1 and T_2 , and what is the variance of such a combination? [7]

Question 3

- (a) Explain the following with suitable example:

- (i) Simple and composite hypothesis
- (ii) Types of errors in hypothesis testing
- (iii) Critical region and P-value
- (iv) Power of a test

[7]

- (b) State weak law of large numbers. Examine whether the weak law of large numbers holds for the sequence $\{X_k\}$ of independent random variables defined as follows:

$$P[X_k = \pm 2^k] = 2^{-(2k+1)}, \quad P[X_k = 0] = 1 - 2^{-2k} \quad \text{or not.}$$

[7]

Question 4

- (a) Twenty people were attacked by a disease and only 18 survived. Will you reject the hypothesis that the survival rate, if attacked by this disease, is 85% in favour of the hypothesis that it is more, at 5% level of significance? (You may use large sample test) [Given that $\sqrt{0.006375} = 0.0798$, and $Z_{0.05} = 1.645$]

[7]

- (b) Describe Chi-square test of independence of attributes and its assumptions. Also, show that:

$$\chi^2 = \left[\sum_{i=1}^n \left(\frac{O_i^2}{E_i} \right) \right] - N$$

where notations are having their usual meaning.

[7]

Question 5 What is difference between dependent and independent sample t-tests? Write assumptions for t-test to make inferences about significance difference between two population means. A drug is given to 8 patients and the increments in their blood pressure are recorded to be 4, 0, 7, -2, 0, -3, 2, 0. Assume that increment in their blood pressure follows normal distribution. Is it reasonable to believe that the drug has no effect on the change of blood pressure at 5% level of significance?

[Given that $\sqrt{10.57} = 3.25$, $\sqrt{2} = 1.4142$ and $t_{7,0.025} = 2.365$]

[14]

Question 6 Discuss sign test. The following data give the milk production (in thousand kg) in full cream by 36 different dairies:

17	15	20	29	19	19	22	25	27
24	20	17	6	24	14	15	23	24
19	23	28	19	16	22	24	17	20
19	10	23	18	31	13	20	17	24

Use sign test to test that median production of milk in dairies is 21.5 thousand kg at 5% level of significance.

[14]

Question 7 Describe any **THREE** of the following:

- (a) Kolmogorov-Smirnov test and median test [4.66]
- (b) F-test and its applications [4.66]
- (c) Central Limit Theorem with its applications [4.66]
- (d) Large sample tests for variables [4.66]